

The Charged Coupling: The Infrared Catastrophe as a Power Law, Sector Pumping, and the Memory-Charge Relation

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****Abstract.**** The measurement gate proved that compact bilinear couplings preserve the memory superselection sector and located sector dynamics at source-type couplings; the covariance audit directed all interactions to the representable channels. This note walks through that doorway and completes the interaction taxonomy, exactly. Linear source couplings integrate in closed form: the S-matrix is a Weyl operator (time-ordering contributes only a phase). Current-type sources, with compact smearing, are implementable and sector-preserving. The critical case is the **charged** coupling — a source for the field ϕ with total charge $Q = \int g \neq 0$ — whose current-frame smearing is a primitive with an infinite plateau of height Q : ****the coupling operator is the memory operator of the soft-sector paper****, and its proven divergent variance makes the S-matrix non-implementable in the vacuum sector. Regularizing by marching the compensating charge to distance L yields three theorems, each machine-verified. ***(1)*** The infrared catastrophe as a closed-form power law: $\|KG_L\|^2 = 2Q^2 \ln L + O(1)$ — measured slope 0.3943 against $2Q^2 = 0.39426$, with exact charge-squared scaling (ratio 4.000 under $Q \rightarrow 2Q$) — so the vacuum-persistence amplitude vanishes as

$$|\langle \Omega, S_L \Omega \rangle| \sim L^{-\lambda^2 Q^2},$$

the exponent proportional to (coupling $\times$ charge)², the soft-factor shape of gauge theory arising here from the archive's own memory-variance theorem. ***(2)*** All local observables converge — the source-region current expectation is +0.575183 identically across the entire ladder, and the plateau region between separated charges carries exactly zero current — while the global amplitude dies: the infraparticle resolution, with finite physics and the vacuum-sector unitary lost to a sector map. ***(3)*** The limiting state lies in the memory sector $m = -2\pi\lambda Q$ — verified to all computed digits — a **memory-charge relation**, linear in the source: the model's own soft theorem. The taxonomy closes: bilinear couplings preserve, current-type sources preserve, charged sources pump memory $-2\pi\lambda Q$; and the question of **which** charges arise is handed, by name, to the multiplicity structure of the audit's third channel.

1. Through the doorway

Two results framed this note. The measurement gate's theorem: compact bilinear couplings cannot pump memory, and sector dynamics require sources that couple non-derivatively. The covariance audit's map: interactions enter only through representable channels — states, sources, patterns. The charged source is the minimal representative of the second channel, and it is exactly solvable.

2. Theorem 1: exact S-matrices and the critical case

For a linear coupling $H_{\text{int}}(t) = \int J(t, u) j(u - t) du$ with compactly supported J , currents at different arguments commute to c-numbers, so time-ordering contributes a phase and the S-matrix is exactly a Weyl operator, $S = e^{i\varphi} W(h)$, with h the source integrated along characteristics. Two cases split the physics. **Current-type sources:** h compact; S is implementable, the radiated wave $\ell = 2\pi h'$ has $\int \ell = 0$, and the sector is preserved — sources of the current cannot pump the current's zero mode. **Charged sources:** a source for the field, $S = e^{-i\lambda\phi(g)}$ with charge $Q = \int g \neq 0$, has current-frame smearing $G = \int^u g$ — a primitive climbing to the permanent plateau Q . This G is the memory window of the soft-sector paper, and by that paper's theorem $\phi(g)$ has logarithmically divergent variance: the coupling operator does not exist on the vacuum sector, and neither does its exponential. The obstruction is not a pathology; it is the superselection structure announcing itself in the dynamics. ■

3. Theorem 2: the infrared catastrophe as a closed-form power law

Regularize by total neutrality at finite range: place the compensating charge at distance L , so G_L is compact with plateau Q over length $\sim L$. Then, by the memory-variance law with plateau height Q ,

$$\|KG_L\|^2 = 2Q^2 \ln L + O(1) \quad \implies \quad |\langle \Omega, S_L \Omega \rangle| = e^{-\lambda^2 \|KG_L\|^2/2} \sim C L^{-\lambda^2 Q^2}.$$

Verified: successive-doubling slopes of $\|KG_L\|^2$ are 0.3943, 0.3939, 0.3925, 0.3872 against the prediction $2Q^2 = 0.39426$ ($Q = 0.443994$ for the unit test bump), and quadruple exactly under $Q \rightarrow 2Q$ ($1.5773/0.3943 = 4.000$); at $L = 320$ the persistence is 2.55×10^{-1} for charge Q and 4.25×10^{-3} for $2Q$. The exponent $\lambda^2 Q^2$ — coupling times charge, squared — is the structural shape of the soft factors of gauge theory, obtained here as a corollary of the archive's memory-variance theorem rather than from diagrammatics.

4. Theorem 3: sector transition and the memory-charge relation

Two limits coexist, and their contrast is the physics. ****Local convergence:**** every local expectation converges — the current smeared at the source region gives $+0.575183$ ***identically*** across the ladder $L = 20 \rightarrow 320$, and a smearing in the plateau between the separated charges gives exactly zero (the classical wave lives at the charges; between them the current vanishes — the geometry read back from the numbers). ****Global death:**** the vacuum-persistence amplitude vanishes as the power law of §3. The resolution is the infraparticle one, here as theorem rather than lore: the physics is finite, but the limiting state is not in the vacuum folium — it is the shifted quasi-free state with local memory density integrating to

$$m = -2\pi\lambda Q,$$

verified to all computed digits (-2.789695 against $-2\pi\lambda Q = -2.789695$), lying in the memory- m superselection sector, disjoint from the vacuum sector by the soft-sector paper.

The compensating anti-memory has been radiated beyond every bounded window. The relation $m = -2\pi\lambda Q$ — memory linear in charge — is the model's soft theorem, and the standard dressed-state formalisms (the Faddeev–Kulish tradition, cited by name) are its established language.

5. The taxonomy, closed — and the pattern question, opened

$\begin{array}{lll} \text{compact bilinear coupling} & \longleftrightarrow & \text{sector-preserving (measurement gate, exact)} \\ \text{linear current-type source} & \longleftrightarrow & \text{sector-preserving (Theorem 1)} \\ Q \neq 0 & \longleftrightarrow & \text{sector-pumping, } m = -2\pi\lambda Q \text{ (Theorems 2-3)} \end{array}$

Every interaction class the operator theory admits now has its sector verdict. What the taxonomy cannot say is *which* charges and couplings nature's version of this model contains — and that is precisely the audit's third channel: the sedenion-level multiplicity structure as the source of coupling patterns and selection rules. That thread — the generation algebras assigning which currents carry which charges — is the named next question of the program.

6. Scope

Five items. (i) The statement is per-generator chiral; assembled over the sphere, directional charges $Q(\hat{n})$ pump a memory profile $m(\hat{n}) = -2\pi\lambda Q(\hat{n})$ — the supertranslation-shaped label of the soft-sector paper, engaged direction by direction. (ii) Memories compose additively across successive charged scatterings; the sector label is an abelian charge of the dynamics. (iii) Back-reaction of the sector label on subsequent dynamics beyond leading order is deferred. (iv) The gravitational analogue of the memory–charge relation is spin-2 and outside this note's claims. (v) The generation/selection-rule thread is next.

References

- Companion papers: *The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering; The Soft Sector: Memory as a Superselection Label and the Harmonic-Number Structure of the Zero-Mode Kernel; The First Law on the Net: The Stress Tensor, the Wedge Modular Hamiltonian, and the Relative Entropy of Coherent Excitations; The Covariance Audit: Why the Native Deformation Cannot Ride the Boosts, and Where Interactions Must Enter.*
- Fewster, C. J. & Verch, R., "Quantum fields and local measurements," *Comm. Math. Phys.* 378 (2020), 851–889.
- On dressed states and the infrared representation of charged sectors: the Faddeev–Kulish tradition (cited as tradition; no single-theorem attribution intended).

Created in collaboration with Claude (Anthropic). All conclusions and any errors remain the author's responsibility.